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Research article

Spontaneous eye blink rate: An index of dopaminergic component of sustained attention and fatigue

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ABSTRACT

Blink rate is a behavioral index highly correlated with frontostriatal dopaminergic activity. The present research was aimed at studying the modulation of spontaneous blink rate in function of the increasing attentional load induced by the Mackworth Clock Test. Since blinking interferes with sensory processing, we expected a decreasing blink rate with increasing attentional demand. Three tasks of 7-min each and different difficulties were administered: the Mackworth had a red dot moving in a circle with intervals varying from 500 ms, 350 ms to 200 ms, corresponding to increasing task difficulty. Participant had to detect the rare jumps of one position by the red dot (targets). The blink rate was recorded from thirty-three female students starting from vertical oculogram recording of the right eye. The time course of blink rate across the 7-min task was also analyzed to test the hypothesis that fatigue arises also during brief tasks depending on the difficulty level. Results showed that the Hard task (200 ms dot intervals) was associated with greater percentage of missed targets, faster response times and smaller blink rates with respect to the Medium and Easy ones. Analysis of the time course within the task revealed an increase of blink rate, indexing larger fatigue, starting in the 4th minute, independent from the difficulty level. In addition, trial-by-trial analysis showed that under strong attentional demand dopamine-related blink activity was inhibited throughout the whole task. Results point to the use of blink rate as an ecological index of dopaminergic component of attentional load and fatigue and revealed how human attention drops after relatively brief intervals of about 4 min.

1. Introduction

Spontaneous eye blink refers to the brief closure of eyelids which occurs without any stimulation and without volition (McMonnies, 2010). It should not to be confused with corneal reflex, which is triggered by an external tactile stimulation of the eye, and serves as a defensive mechanism to preserve eye integrity. There is consistent evidence indicating that the frequency of spontaneous eye blink depends on the dopaminergic activity in the central nervous system (Jongkees and Colzato, 2016). Recent findings from an animal model of spontaneous blinking activity suggest that the trigeminal spinal complex is involved in this process (Kaminer et al., 2011). This nucleus receives input from the trigeminal nerve which carries sensory information on the state of the cornea, and it is modulated by basal ganglia through connections that involve the nucleus raphe magnus and the superior colliculus. The last pathway explains the relationship observed between dopamine levels and spontaneous eye blink rate (EBR). Experimental manipulation of the dopaminergic activity in healthy subjects revealed that frequency of spontaneous blink raises in parallel with the increase of dopaminergic activity (Jongkees and Colzato, 2016). Further support for this association has been found in studies of individuals with altered dopaminergic function, such as patients affected by Parkinson's Disease (PD) or schizophrenia (Karson et al., 1984; Karson et al., 1990). The former show a pattern of diminished blinking activity, due to the depletion of dopamine in basal ganglia, while schizophrenics, affected by an excess of mesocorticolimbic dopamine, are characterized by increased blinking. It is interesting to note that these pathological conditions are also characterized by attentional dysfunction (Green, 2006; Nieoullon, 2002), suggesting a possible role of these pathways in attention. Dopamine depletion that characterizes PD patients has been linked to decreased performance in neuropsychological testing of attention and executive functions, in which these patients consistently perform with an increased cognitive inflexibility and an inability to shift their attention to relevant information in order to comply with environmental demands (Nieoullon, 2002). On the other hand, patients affected by schizophrenia, typically affected by severe impairment of attention orienting, show a strong inability to avoid processing irrelevant information (Nieoullon, 2002): for these

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patients every stimulus, also insignificant, becomes relevant, a phenomenon which has been interpreted as one of the basic mechanism of the positive symptoms which characterizes this condition.

Since eye blink measurement is relatively easy and non-invasive, this behavioral index has been used to investigate the role of dopamine in several neurological and psychiatric conditions (Ebert et al., 1996; Karson et al., 1984, 1990; Kojima et al., 2002; Ladas et al., 2014), as well as in the investigation of normal behavior and personality (Barbato et al., 2012; Chermahini and Hommel, 2010; Colzato et al., 2009a; Colzato et al., 2009b; Dreisbach et al., 2005; Müller et al., 2007). Interestingly, attention is modulated by several neurotransmitters, including those involved in arousal/consciousness, that are acetylcholine, noradrenaline, serotonin and dopamine (Robbins, 1997). Thus, among the many neurotransmitters involved in attentional modulation, blink rate may serve as an optimal noninvasive index of dopaminergic component of attention.

In resting conditions, human participants blink with a frequency that ranges between 10 and 25 blinks/min. This measure is characterized by a great inter-individual differences (McMonnies, 2010). Therefore, the frequency of eye blinks is not exclusively a function of eye-dryness, but instead is modulated by several other variables, including the psychological ones. One of the most important variables is the extent of attention allocated to perform a task, a variable usually defined cognitive load. Past classic research (Drew, 1951; Poulton and Gregory, 1952) showed that during visual tasks blink rate was modulated by the difficulty of the task, with an inhibition of blink mechanisms in response to increased task demand. The explanation suggested for this effect was that blink inhibition occurs to avoid the loss of incoming information. Indeed, each blink breaks the continuity of the visual sensory stream (Volkmann et al., 1980). This interpretation has been supported by several findings showing that blink is inhibited when participants are engaged in tasks that require processing of visual information, e.g. when performing motor action under visual guidance or when reading (Cardona et al., 2011; Carpenter, 1948; Holland and Tarlow, 1972; Martin and Carvalho, 2015; McIntire et al., 2014; Oh et al., 2012).

In addition, this mechanism appears to be not restricted to visual tasks, extending to other situations in which visual processing is not involved. Holland and Tarlow (1972), using digit span and a mental arithmetic task, showed that participants' blink rate was reduced when the difficulty of the task (and thus the extent of allocation of attentional resources) increased. Oh et al. (2012) found the same effect in a task in which participants were required to listen to a series of tones, with varying difficulties, and had to report their number.

Nevertheless, there is another factor which has been consistently associated with blink modulation, that is the time spent on performing a task. The time-on-task effect has been linked to diminished vigilance, greater distractibility and lower attention, a cluster of phenomena which is sometimes referred to as mental fatigue (Stern et al., 1984). This relation is extremely relevant due to its practical applications, and there is a growing body of evidence, coming from the field of applied psychology and ergonomics (Martin and Carvalho, 2015), supporting the idea that blink rate increase is an indicator of mental fatigue.

In a study performed by Carpenter (1948) participants had to perform the Mackworth Clock Test (Mackworth, 1948), a task designed to assess the sustained vigilance of the participant for 2 h: results showed an increase of blink rate across time and throughout the experiment. Fukuda et al. (2015), showed that blink frequency increased during a monotonous memory task, and McIntire et al. (2014) used an ecological paradigm to demonstrate a pattern of increasing blink rate in participants performing a simulation of air-traffic control.

It remains unclear whether the effects of fatigue across time and difficulty (meant as increased attention allocation) operate on the modulation of blink rate through a common mechanism, or instead they exert their effect independently. The present study sought to contribute to this topic by introducing a paradigm in which both the difficulty of the task, and the time spent by participants on it, were introduced and analyzed. Using a vigilance task paradigm with different levels of task difficulty, we tested the hypothesis that blink rate changes as a function of task difficulty and duration independently, or, alternatively, that the two variables interact.

2. Method

2.1. Participants

Thirty-three female students (mean age = 22.9, range = 21–26 years) participated in this study in exchange of credit course. All participants were healthy and had normal or corrected to normal vision. The study was approved by the Psychology Ethics Committee, University of Padova. The investigation has been conducted according to the principles expressed in the Declaration of Helsinki. Before starting the recording session, the participant was given a general description of the experiment and the procedure in order to obtain the approved informed consent.

2.2. Stimuli and task

An adapted version of Mackworth Clock Test (MCT) (Mackworth, 1948) was used in the present study in order to investigate the relationship between spontaneous blink rate and attentional performance. Stimuli consisted in 100 white dots (radius = 2.6 mm) arranged to form a circle (radius = 105 mm) displayed at the center of a computer screen. In each trial, a red dot was displayed moving across the circle, jumping from one position to the following at a constant rate. For the purpose of the experiment, we decided to manipulate the difficulty of the task by changing the speed of the red dot moving across the circle, which was 200 msec (Hard condition), 350 msec (Medium condition) or 500 msec (Easy condition). Trials duration was 7 min. The task consisted in detecting when the red dot jumped a position, i.e. a white dot, by pressing the space button on the computer keyboard. Target events (jumps) occurred at random positions and represented 1% of total dot moves. Behavioral measures derived from the task were the Response times to target events and the percentage of missed targets (target that participants failed to detect). Each participant was administered three Mackworth tasks of 7 min duration each, with a pause of few minutes between tasks, the order of the Mackworth tasks varying for difficulty (Easy, Medium, Hard) were randomized and balanced across participants.

2.3. Eye blink acquisition and preprocessing

Spontaneous eye blinks were identified through vertical electroculogram collected by means of two electrodes placed above and below the right eye. Signal was amplified with a gain of 4000 and online filtered with a time constant of 10 s and a low pass filter set at 80 Hz. A high pass filter set at 0.5 Hz was applied offline in order to remove slow oscillations due to movement artifacts. Eye blinks were defined as a peak of positive voltage change exceeding the threshold of 100 microvolts in a time window of 500 msec (Colzato et al., 2009b). In order to assess the evolution across time of blink rate, each trial was divided in 7 blocks of one-minute duration and mean blink rate was computed for each block. Furthermore, to control for the large inter-individual differences in spontaneous blink rate (Barbato et al., 2000), the ratio between EBR collected during the task and the EBR collected during the baseline was computed.

We were also interested to explore trial-by-trial variation in eye blinks, a novel approach that allowed us to investigate phasic changes in dopaminergic activity in more detail (Rac-Lubashevsky et al., 2017; van Bochove et al., 2013). Therefore, for each trial (i.e. dot presentation) within the three experimental conditions, we also analyzed the occurrence/not occurrence of a blink.

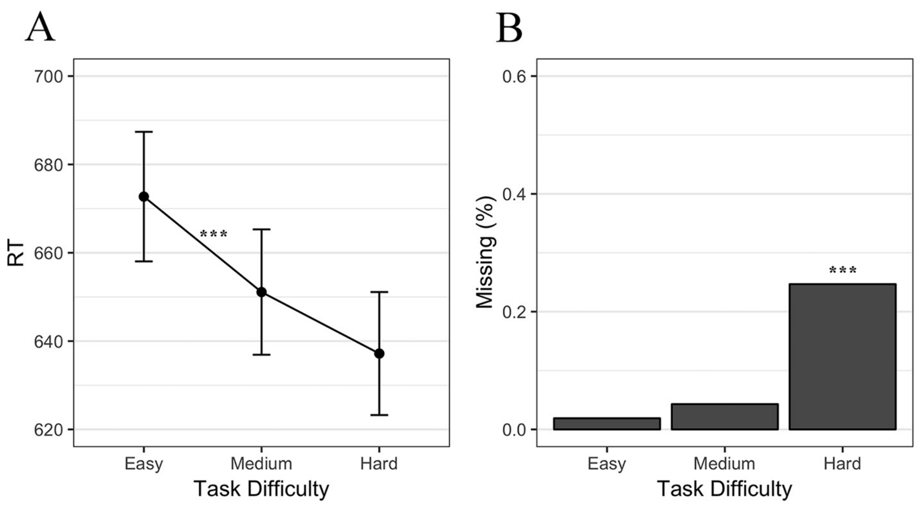

Fig. 1. Effect of Task Difficulty on Response Times to target hits (a dot jumping a position), on left panel (A), and on percentage of Missing targets, right panel (B). *** indicates significant post-hoc effects with $p < 0.05$.

2.4. Procedure

Upon arrival in the laboratory, participants received a general description of the experiment and provided their informed consent. Then, EOG electrodes were attached and participants performed a practice session, with the supervision of the experimenters, in order to acquaint with the task. At the end of the practice trial, the experimenters left the room and the experimental task begun. The order of the tasks was counterbalanced across the participants. Before starting the experimental session participants were asked to fixate a cross presented at the center of the screen for 3 min, in order to collect a baseline measure of their blinking rate. Since blink frequency increases in the evening (Barbato et al., 2000), data were not collected after 5 p.m.

3. Statistical analysis

Independent variables were task Difficulty and Interval, dependent variables were percentage of Blink rate with respect to the baseline, Response times (RT) to the dot jumps and percentage of Missed Targets (MT). For the analysis of EBR and RTs, linear mixed-effects (LME) models were used (Baayen et al., 2008). LME models represent an extension of classical linear models which allow to encompass standard analytical approaches (e.g. *t*-test, ANOVA, ANCOVA) within a more flexible framework characterized by the possibility to include in the model not only fixed-effects, namely the expected source of variation that one is interested to test, but also random-effects, that are sources of variability not explicitly manipulated but nevertheless important in explaining the observed data. Inter- and intra-individual variabilities represent examples of this random source of variability usually encountered in within-subject experimental designs, that are more effectively taken into account using LME models compared to a more classical analysis such as repeated-measures ANOVA (Bagiella et al., 2000; Boisgontier and Cheval, 2016). In addition, LME models have the advantage to easily handle unbalanced designs and missing observations, that in classical Repeated-Measures-ANOVA usually lead to subject exclusion, or force to rely on imputation of the missing observations. After removal of outliers in the response time (RT > 1500 msec or < 100 msec), a linear mixed model was fitted to RT data including Difficulty as fixed effect and Subject as random effect. Fulfillment of assumptions (homoscedasticity and approximate normal distribution of residuals) were assessed by graphical inspection of model residuals by means of residual plot (residuals plotted against fitted data) and quantile-quantile plots (residuals plotted against a random sample from a normal distribution). This approach allowed to immediately recognize gross departures from normality that could inflate the results, overcoming the low sensibility of formal statistical tests like Shapiro test (Anscombe, 1973; Caudek et al., 2015). An F-test with Kenward-Rogers approximation for degree of freedom was carried out to assess significance of the fixed effect (Luke, 2016). Significant post-hoc effects were further explored by using the Tukey test for pairwise comparisons.

A linear mixed-effects model was fitted also to the blink rate data by including two fixed effects (Difficulty and Interval) and one random effect (Subject). An F-test with Kenward-Roger approximation for degree of freedom was carried out to assess significance of the two main effects and their interaction (Luke, 2016). Significant effects have been further explored by using Tukey test for pairwise comparisons.

Due to the very skewed distribution of the dependent variable, the effect of trial difficulty on missed targets (MT) was analyzed with the non-parametrical Friedman test. Post-hoc comparisons were carried out by using pairwise Wilcoxon tests and Bonferroni correction.

In order to investigate trial-by-trial variation in blink pattern, we analyzed the probability of the occurrence of a blink for each trial as a function of Difficulty and Time using a mixed-effect logistic regression with Subject as random effect.

All the statistical procedures were carried out using R (version 3.3.2) (R Core Team, 2016) and the following packages: lme4 (Bates et al., 2015), lmerTest (Kuznetsova et al., 2016), lsmeans (Lenth, 2016), dplyr (Wickham and Francois, 2016) and ggplot2 (Wickham, 2009).

4. Results

Analysis of RTs (Fig. 1A) revealed a significant effect of task Difficulty ($F_{(2,1101)} = 8.56$, $p < 0.05$). Post-hoc comparisons showed that participants reacted faster to the target stimuli during the Hard task compared to the Easy ($t_{(1101)} = -4.13$, $p < 0.05$) one but not compared to the Medium ($t_{(1101)} = -1.79$, n.s.) task. In addition, RTs were faster for the Medium compared to the Easy condition ($t_{(1101)} = 2.39$, $p < 0.05$).

A significant effect of task difficulty was found for Missing Targets ($\chi^2_{(2)} = 47.7$, $p < 0.05$). Percentage of missing was larger for the Hard condition compared to the Easy ($W = 1029.5$, $p < 0.05$) and the Medium ($W = 954.5$, $p < 0.05$) ones (Fig. 1B).

Analysis of blink rate showed a significant effect of task Difficulty ($F_{(2608)} = 7.004$, $p < 0.05$) and of time Interval ($F_{(6608)} = 3.803$, $p < 0.05$), without interaction between the predictors ($F_{(12608)} = 0.47$, n.s.). Post-hoc analysis on the Difficulty effect (Fig. 2) revealed that eye blinking was reduced in the Hard condition compared to the other two (Hard-Medium contrast: $t_{(608)} = -3.33$, $p < 0.05$, Hard-Easy contrast: $t_{(608)} = -3.13$, $p < 0.05$).

Concerning the evolution of blink rate across Intervals (Fig. 3), post-hoc analysis showed that EBR increased as a function of time, with larger EBR in the fourth ($t_{(608)} = -3.81$, $p < 0.05$), sixth

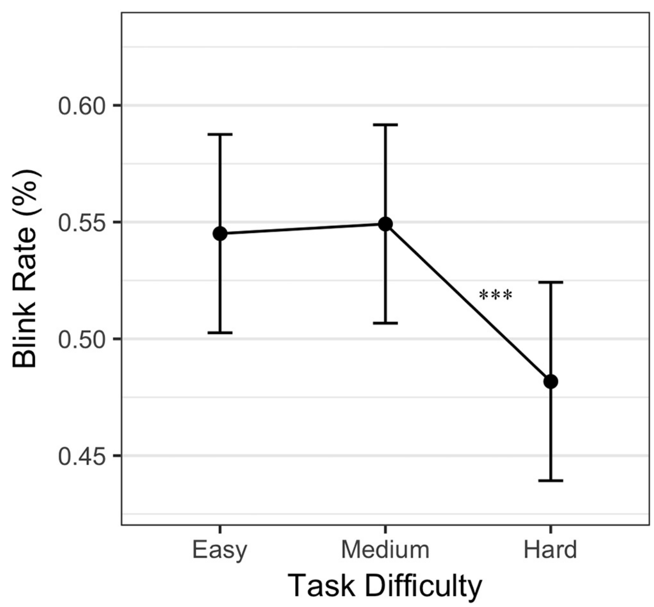

Fig. 2. Effect of Task Difficulty on Blink Rate, computed as percentage with respect to the baseline blinks. *** indicates significant post-hoc effects with $p < 0.05$.

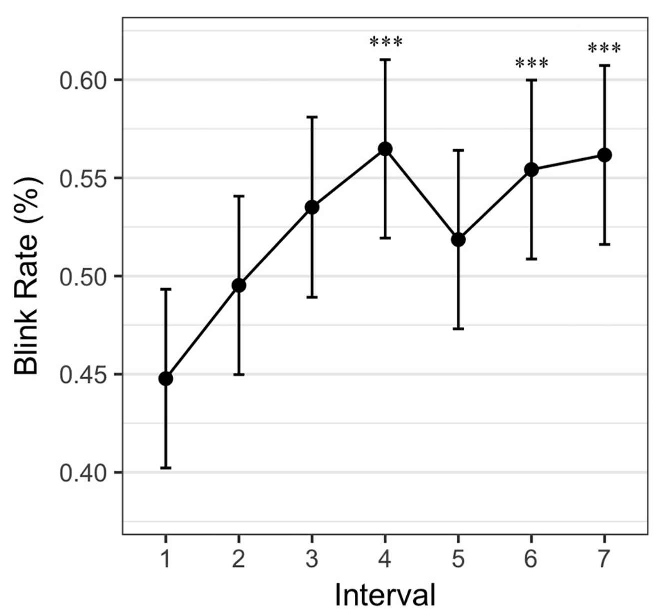

Fig. 3. Time course of blink rate (percentage) during the 7-min duration task (the three task difficulties are collapsed). This was broken into seven 1 min Intervals. *** indicates significant post-hoc effects with $p < 0.05$.

($t_{(608)} = -3.45$, $p < 0.05$) and seventh ($t_{(608)} = -3.69$, $p < 0.05$) intervals compared to the first.

Analysis of trial-by-trial blink probability revealed a main effect of trial Difficulty ($\chi^2_{(2)} = 1145.89$, $p < 0.05$), a main effect of Trial ($\chi^2_{(1)} = 20.22$, $p < 0.05$) and a significant Difficulty by Trial interaction ($\chi^2_{(2)} = 7.41$, $p < 0.05$). Analysis of the coefficients showed that the probability of a blink during dot switch-on decreases progressively from the Easy condition ($z = -18.31$, $p < 0.05$), to the Medium condition ($z = -8.74$, $p < 0.05$) to the Hard ($z = -18.34$, $p < 0.05$). Fatigue emerged across time as the probability of blinking during dot lighting increased at the end of the task in both the Easy ($z = 2.12$, $p < 0.05$) and the Medium ($z = 4.44$, $p < 0.05$), but not in the Hard condition ($z = 1.84$, n.s.) (Fig. 4).

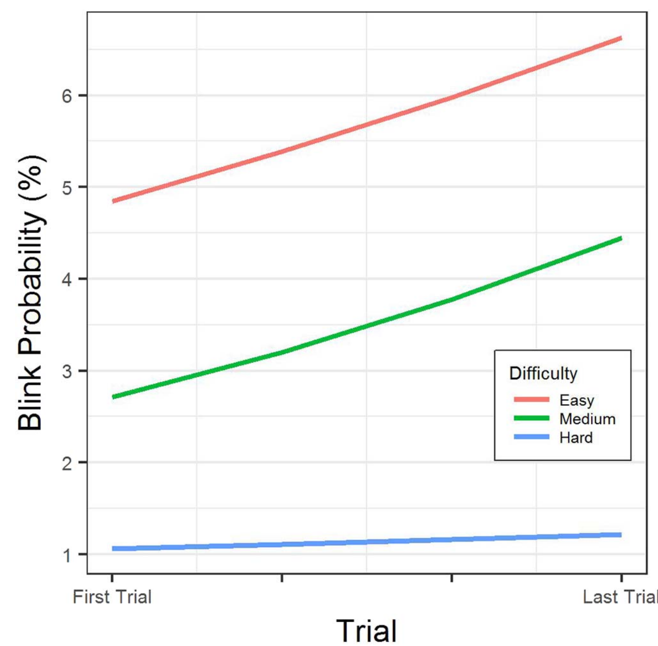

Fig. 4. Regression lines indicating the time course of fatigue within each task, computed on a trial by trial basis. The dependent variable is the probability to find a blink during each red dot switch-on, for the three task difficulties.

5. Discussion

The aim of the present study was to use a vigilance task to investigate the effect of task difficulty and the time course of spontaneous blink rate on attention modulation. Behavioral results confirmed our initial hypothesis, showing that participants were faster when responding to target presented with higher rate and made more missing targets compared to the easiest conditions. This suggests that the task Difficulty modulated properly the attention allocation required to perform the task. Concerning the blink rate modulation, we advanced two alternative predictions, one suggesting an interacting effect of task Difficulty and Time and another suggesting an independent effect of these factors. The former prediction was based on the hypothesis that in a difficult task, with greater attentional demand, the effect of time would be larger compared to a task requiring fewer attentional resources. The latter prediction instead suggested that the two factors are independent. Analysis of blink rate modulation across seven one-minute intervals clearly support the second hypothesis, showing that blink rate was modulated by task difficulty as well as by interval, without any interaction between the two. The greater blink inhibition observed in response to the task with greater difficulty, together with the behavioral results, confirms the validity of blink rate as an index of attention. Indeed, the increased attention allocation required by this condition is associated with a marked suppression of blinking activity. The three behavioral indices registered showed a different sensitivity to task difficulty: blink rate and number of missing targets showed a clear effect of the most difficult task compared with the Medium and Easy ones: the two latter were equivalent showing that the difficulty in the Mackworth does not increase linearly with decreasing interval between the dots. For this reason the Easy and Medium levels were equivalent. Response times instead reflected the speed of the dot and therefore task difficulty: with increasing dot speed the response times became increasingly fast. This is an interesting effect that shows how the participant spares resources and tends to respond faster only when the intervals become short and force him to speed the average response. Concerning the blink rate suppression, it represents a crucial mechanism that aims at preserving the continuity of the information stream (Volkmann et al., 1980), thus EBR was drastically diminished in conditions demanding high attentional effort. This observation is in line with the view that phasic changes in dopaminergic activity modulate the allocation of attentional resources for stimulus processing leading to a marked reduction in blink rate necessary for the optimal execution of the task. Neuroimaging studies reporting the involvement of frontal regions in spontaneous blinking (Hyo et al., 2005), support the idea that the decrease of dopaminergic activity could be mediated by a top-down influence of prefrontal cortex, which is directly involved in the complex coordination of attentional processes. The analysis also revealed that blink rate changed across the time, with higher blink frequency at the end of the task compared with the beginning, a result which suggests an increase of fatigue in the second half of the 7-min task. The effect was observed in all tasks independently from the difficulty. This result is interesting because it extends the findings of previous experiments that used this measure in tasks of longer durations (Carpenter, 1948; McIntire et al., 2014), showing that this pattern is not limited to long lasting vigilance tasks. Several factors should be considered when interpreting this result, in particular the interaction between the eye physiology, the cognitive mechanisms that modulate spontaneous blinking and the role of dopamine in cognition. Eye blinking is a behavior which primarily serves the purpose of maintaining the eye hydrated, with a thin tear film aimed at protecting the cornea (Al-Abdulmunem, 1999). When the environment requires to focus attention toward a stimulus, this physiological mechanism can be partially abolished, leading to a tradeoff between the need to inhibit the blink for preserving the visual information and the need to maintain eye hydration. An example of this tradeoff occurs during reading, in which there is a temporal adjustment of blinking activity represented by the production of a burst of blinks at the end of a sentence (Cardona et al., 2011). This phenomenon could be interpreted as the reflection of rebound mechanisms aiming at restoring a tonic dopaminergic activity, following the initial investment of attentional resources. Moreover, our results suggest a more complex mechanism since we found a significant difference also between the first and the fourth intervals. This pattern could be interpreted as an evidence of an ultradian (short-duration rhythm) component in attention indexed by EBR. Similar observations have been previously reported in other studies, which consistently observed that the performance in an attentional task was characterized by periodic oscillations, with the peak decrement occurring with a period of about 4 min (Conte et al., 1995; Smith et al., 2003). Moreover, an electrophysiological study showed that this periodicity in the performance was linked to increased low frequency EEG activity, especially Delta and Theta, and a concomitant decrease in higher frequencies like Gamma oscillations (Makeig and Jung, 1996). These results have been interpreted as a neurophysiological signature of arousal fluctuations mediating rhythmic decrease in attention and vigilance. Our results provide a new insight on this phenomenon showing that the EBR increment found during an attentional task follows similar ultradian 4-min period. Indeed, optimal cognitive performance does not relate to dopamine levels in a linear fashion, following instead an inverted U-shaped relationship (Cools and D'Esposito, 2011), which means that the best performance is achieved when the dopamine levels are neither too low nor too high. Therefore, these observations could be interpreted as the reflection of a homeostatic process, through which the attentional system progressively compensates for the dopamine inhibition caused by the allocation of attentional resources toward the environment. Thus, an optimal dopaminergic tone should be achieved by balancing the need to limit depletion of all the resources, and the need to interact with the external world.

In order to gather a more fine-grained view of the relationship between EBR and time on task under increasing cognitive load, we also analyzed trial-by-trial variation in eye blinks. This approach allowed a better clarification of the effects of fatigue across time, showing that when considering blink activity at the level of single trial, and not in aggregate fashion (one-min intervals), task difficulty and time do effectively interact. We found an increase of blink occurrence across trials and time under low and intermediate cognitive loads but not during the Hard difficulty condition. This result is interesting because it further adds to analysis carried out by dividing tasks into 7 one-min intervals. Specifically, it shows that in conditions in which attentional demand is strong (Hard task), the blink-related dopaminergic system remains markedly inhibited for the whole task, without showing increased fatigue across time. This may depend on the high-frequency of stimulus presentation which determines a bottom-up stimulus orienting, driving automatic attentional resources and this in turn would lead to strong blink inhibition throughout the whole task. On the other hand, under conditions in which stimuli are presented with slower rhythm, the increased probability of blinking may reflect the interaction between fatigue and top-down attentional control over the task, which is mediated also by dopaminergic frontostriatal circuits. This result may further explain the tradeoff observed in the behavioral performance between speed and accuracy. In the Hard condition participants are faster but also more inaccurate constantly across the whole task, probably because their behavior is driven by bottom-up processes of automatic attentional orienting, a process more resistant to fatigue. By contrast, in the easier tasks there is a greater top-down control which leads to greater accuracy achieved by paying the cost of slower reaction times and increased mental fatigue: this effect would be marked by a relative increase of blink frequency toward the end of the task. Taken together, results support the idea that the time course of blink rate is a general phenomenon which occurs whenever the interaction with the environment requires an active and controlled allocation of attentional resources, and would reflect the balance of the central dopaminergic tone which is disrupted under strong attentional demands. In addition, EBR modulation by task difficulty reflects the strive of the attentional system to minimize the loss of information that is linked to eye closure. In summary, our results achieved by manipulating Mackworth task difficulty increased our knowledge on the association between blink rate and attention, although this role appears to be complex. In this study, using a non-invasive behavioral index we showed how the allocation of attention occurs in tasks with varying difficulties, but also how this index changes over time. The increased blink rate observed across time should be interpreted as an index of arousal decrement and mental fatigue increase. In future studies, it would be interesting to explore how these effects observed in visual tasks can be extended to other sensory modalities, since previous studies already showed that blink rate modulation by task difficulty occurs also in auditory modality (Oh et al., 2012). It would be also interesting to study how individual differences can modulate these effects. According to a recent study, long-term meditation, a practice that is usually associated with greater cognitive flexibility and increased ability to focus attention, allowed meditators compared with controls to have lower resting EBR (Kruis et al., 2016). It is possible that this characteristic prompts a differential recruitment of dopaminergic pathways under strong attentional load. Finally, novel aspects of blink rate could also be investigated by studying pathological conditions involving alterations of dopaminergic activity (Parkinson, schizophrenia, drug addiction, pathological gambling, etc.).

In conclusion, results of this study point to the use of blink rate as a noninvasive behavioral index modulated by dopaminergic component of sustained attention and sensitive also to short-term fatigue. This index can be easily recorded and analyzed, it can be measured in a number of different ways (with eye-tracker, electro-oculogram, videorecording) and in many different ecological conditions (e.g. during driving as an index of fatigue, or visual exploration for product choice, during sport activity and videogames, during emotional movies, etc.).

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